

An Empirical Comparative

Evaluation

Traditional Image Segmentation Algorithms

Prepared By: Wasi Al Safa, Sunan Rabby, Md. Shibbeer Thalha

Submitted To: Utsha Das (RUET CSE)

Introduction & Objectives

Bridging the Gap

Image segmentation acts as a critical bridge between raw pixel grids and high-level semantic object comprehension.

Rigorous Benchmarking

This research investigates the mathematical capabilities and execution properties of 6 classical spatial, edge, and topological extraction systems.



Evaluation Dataset Pool



3,606 Unique Images

Categorized across 7 distinct object profiles featuring balanced mixtures of artificial geometries and complex organic textures.

- ✓ **Cars & Bikes:** Intricate structural borders.
- ✓ **Organic (Dogs/Cats/Horses):** Deformable textures.
- ✓ **Human:** Highly variable silhouettes.
- ✓ **Flowers:** High chromatic contrast.

The 7-Step Pipeline

Traditional vision algorithms are highly sensitive to microscopic environmental perturbations and sensor noise.

- ✓ **Scaling:** Uniform 256x256 resolution.
- ✓ **Luminance:** 8-bit Grayscale conversion.
- ✓ **Noise Suppression:** Median & Gaussian filtering.
- ✓ **Contrast:** CLAHE adaptive equalization.
- ✓ **Refinement:** Morphological Opening/Closing.

1. Original (Resized)



2. Grayscale



3. Median Blur



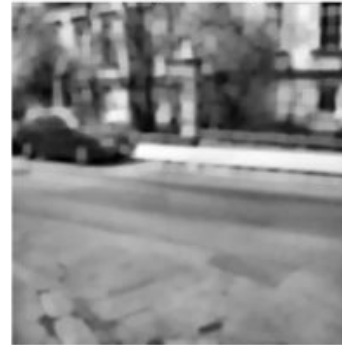
4. Gaussian Blur



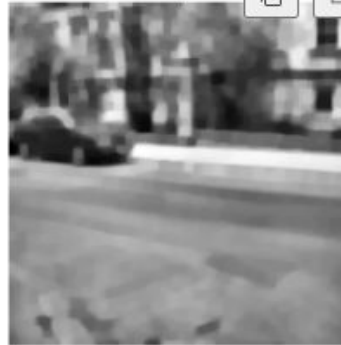
5. CLAHE Contrast



6. Morph. Opening



7. Final Preprocessing



Segmentation Logic: Part 1



Adaptive Thresholding

Computes localized dynamic thresholds based on a Gaussian-weighted mean in a 11x11 window.



Canny Edge Detection

Maps sudden intensity changes using non-maximum suppression and hysteresis thresholds.



Otsu's Thresholding

Calculates optimal cutoffs by minimizing intra-class variance in the global histogram.

Segmentation Logic: Part 2



K-Means Clustering

Unsupervised grouping that clusters pixels into $K=2$ discrete groups via iterative distance optimization.



Region Growing

Deploys from a geometric center seed (128, 128), absorbing pixels within a specific tolerance.



Watershed Segmentation

Floods topological terrain derived from distance transforms to separate touching objects.

Computational Benchmarks

Object Density (%)

$$D_{\text{obj}} = \frac{\sum P_{\text{foreground}}}{\sum P_{\text{total}}} \times 100$$

Indicates under- or over-segmentation leaks.

Contrast Separation (CSR)

$$\text{CSR} = \frac{\mu_{\text{fg}} - \mu_{\text{bg}}}{\sigma_{\text{fg}} + \sigma_{\text{bg}} + \epsilon}$$

Quantifies statistical isolation from the background.

System Evaluation Matrix

Algorithm	Class	Density (%)	Contrast Separation	Exec. Time (ms)
Canny_Edge	Human	5.20	0.19	0.51
Watershed	Cats	30.79	1.31	3.58
K_Means	Cars	62.30	1.32	16.59
Otsu_Threshold	Human	52.48	1.54	0.12
Adaptive_Threshold	Bike	69.62	0.45	0.37
Region_Growing	Dogs	98.57	0.39	0.46

Key Findings & Analysis

Latency Baseline

Otsu is the performance king at **0.10 ms**. K-Means is the slowest at **17.0 ms** due to iterative multidimensional updates.

Boundary Stability

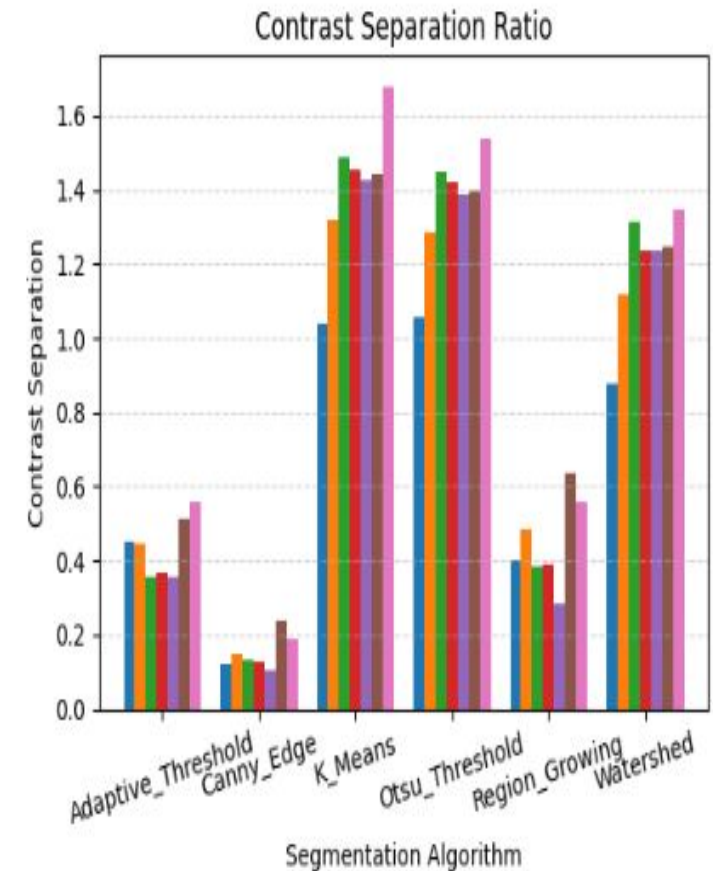
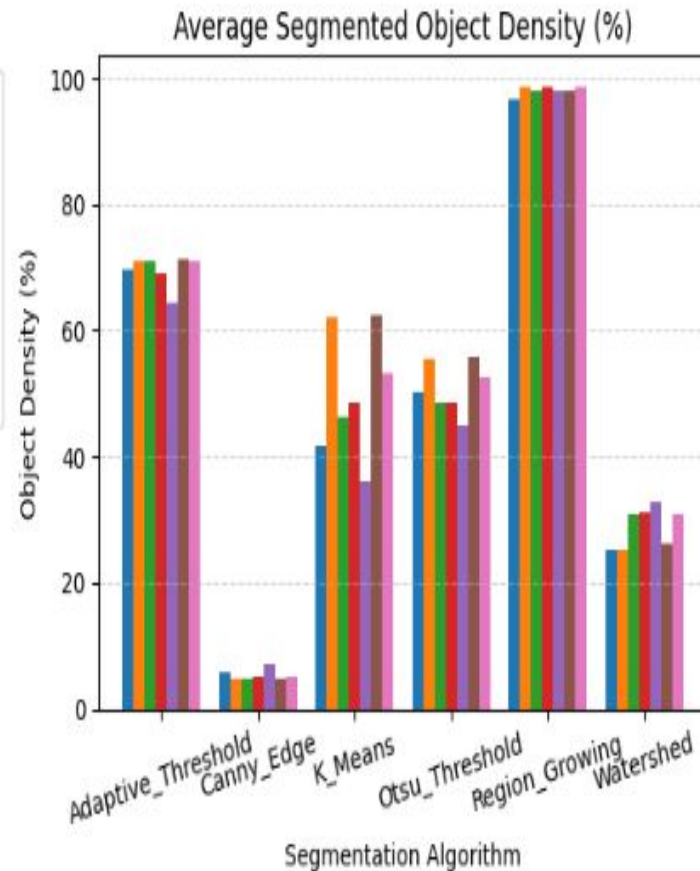
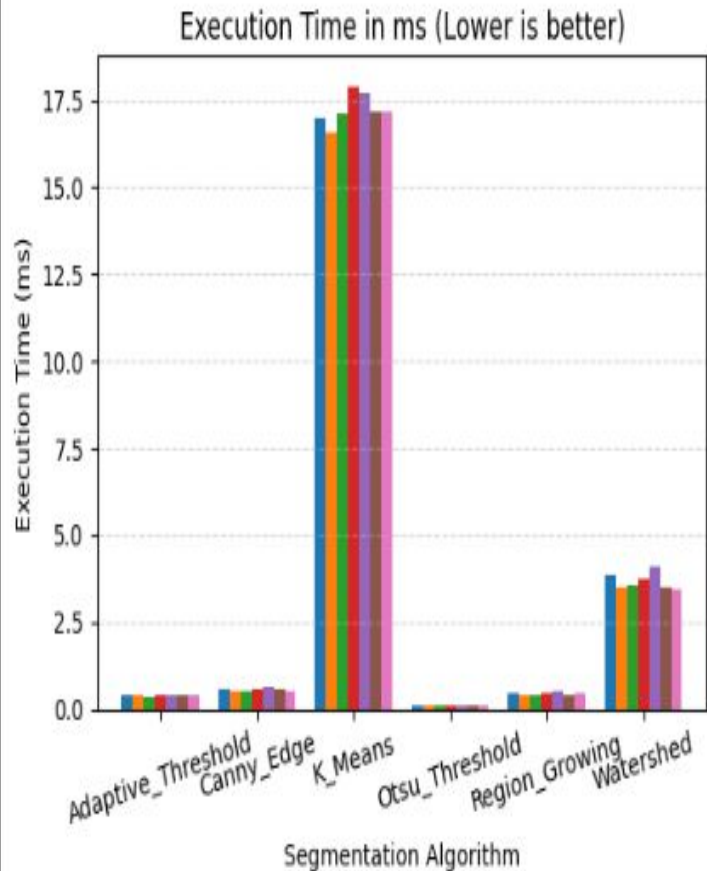
Watershed maintains highly stable spatial counts (25%–32%). Region Growing suffers from massive leakage (>96%).

Isolation Power

K-Means and Otsu achieve CSR > 1.30, proving highly effective at splitting bimodal intensity distributions.

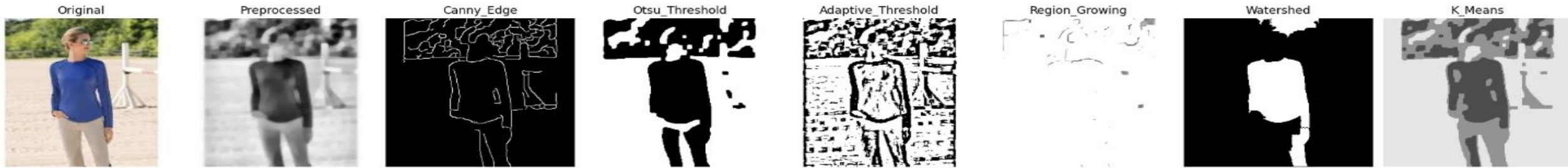
Comparative Performance Charts

Algorithm Performance Comparison Across Categories

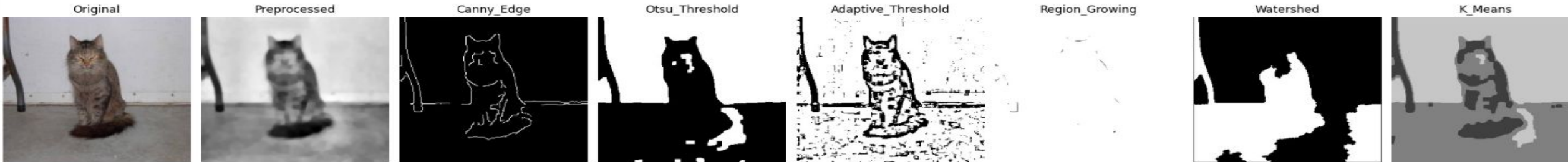


Qualitative Output Matrix

Category: HUMAN | File: rider-107.jpg



Category: CATS | File: cat.58.jpg



Comprehensive cross-categorical qualitative visual comparison across all 7 classes.

Qualitative Output Matrix

Category: DOGS | File: dog.195.jpg



Original

Preprocessed

Canny_Edge

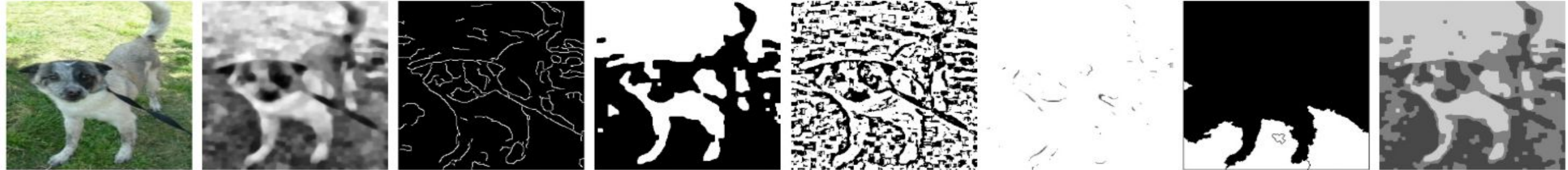
Otsu_Threshold

Adaptive_Threshold

Region_Growing

Watershed

K_Means



Category: HORSES | File: horse-130.jpg



Original

Preprocessed

Canny_Edge

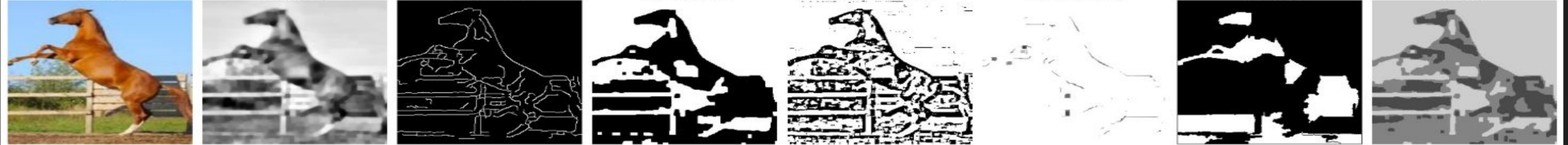
Otsu_Threshold

Adaptive_Threshold

Region_Growing

Watershed

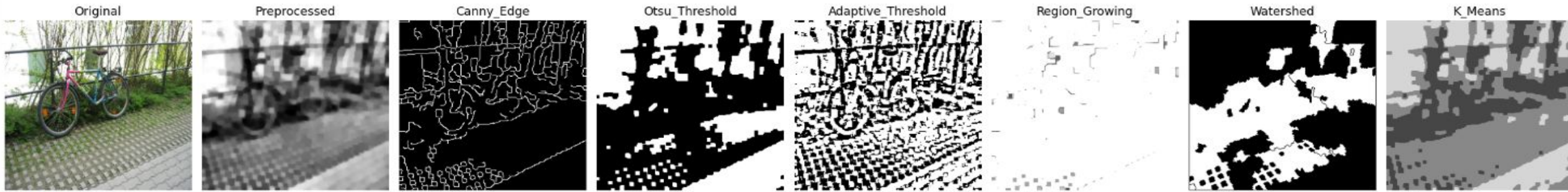
K_Means



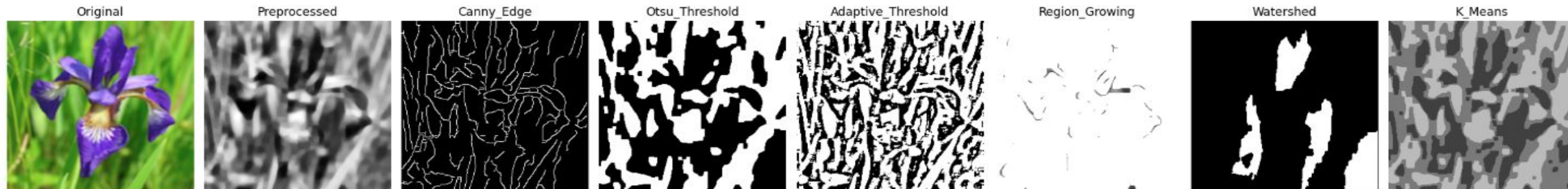
Comprehensive cross-categorical qualitative visual comparison across all 7 classes.

Qualitative Output Matrix

Category: BIKE | File: bike_109.bmp

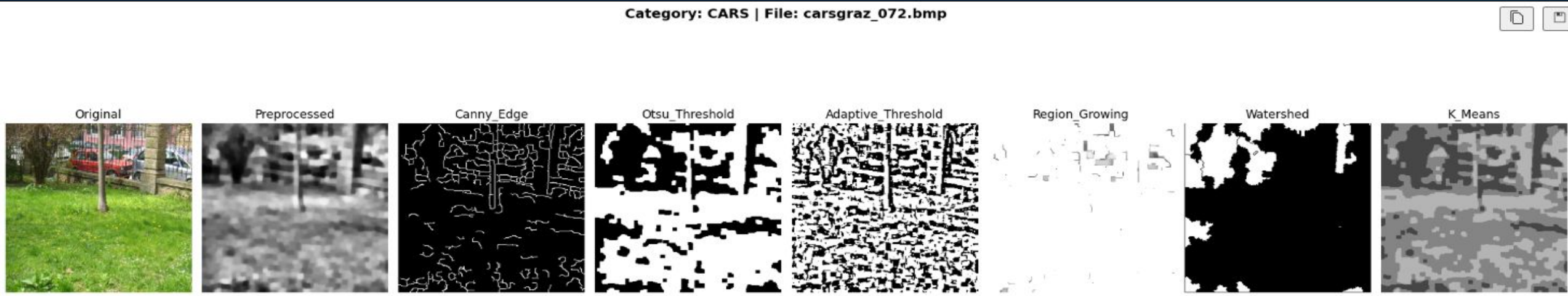


Category: FLOWERS | File: 0156.png



Comprehensive cross-categorical qualitative visual comparison across all 7 classes.

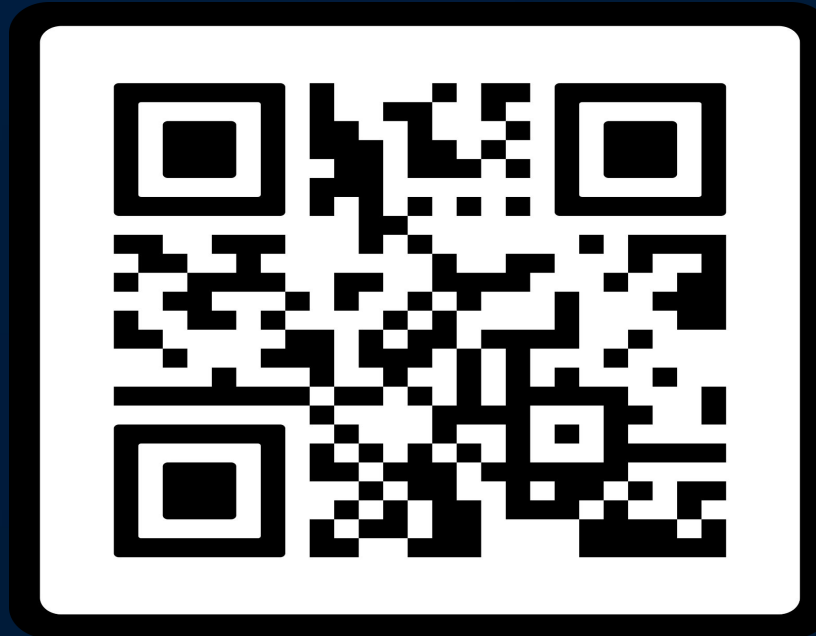
Qualitative Output Matrix



Comprehensive cross-categorical qualitative visual comparison across all 7 classes.

Thank You!

Questions & Discussion



Official Repository:

https://github.com/wasi-coder/DIP_segmentation_project.git

